Multi-Horizon Joint-Embedding Predictive Architectures and Coordinate Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

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***Abstract*—Monitoring systems such as spacecraft and industrial plants must flag sensor faults quickly while staying quiet during normal operation. Detectors that reconstruct raw sensor signals fire often on harmless jitter and drift. Joint-Embedding Predictive Architectures (JEPAs) instead predict compact learned summaries and raise far fewer false alarms, but they still add every summary coordinate into one score, so a few slowly drifting coordinates can raise the alarm threshold and hide faults elsewhere. We replace that single score with a panel of per-coordinate circuit breakers: each coordinate's prediction error at several future horizons receives its own threshold fitted on healthy data, and an alarm fires when enough breakers trip together. We study two versions: Laminar-JEPA-Lite, a compact 128-breaker design recommended for deployment, and Laminar-JEPA, a 224-breaker reference variant. On 118 held-out streams from nine datasets that played no role in design, with checkpoint selection, adaptation, and threshold calibration on separate blocks of healthy data, Laminar-JEPA-Lite raises causal event recall from 52.5 to 63.8 percent over a single-score multi-horizon JEPA, improving on every dataset, and outperforms simpler per-coordinate detectors matched to the same alarm rate. On development streams, latent prediction holds causal false-positive rates near 1 percent, against 22 to 24 percent for reconstruction detectors. Slow ramps lasting four or more context windows remain undetected.**

***Index Terms*—Anomaly detection, Multivariate time series, Representation learning, Self-supervised learning, Statistical process control.**

I. Introduction

Spacecraft, power plants, and industrial machines stream readings from dozens of sensors at once [1]–[4]. A detector watching this telemetry has two jobs: raise an alarm soon after something breaks, and stay quiet the rest of the time. Faults are rare and seldom labeled, so the detector must learn what normal looks like from healthy history alone. And operators lose trust quickly: a monitor that alarms several times a day on healthy data is soon ignored, and an ignored monitor catches nothing.

Gap 1, reconstruction detectors confuse noise with faults: The most common approach trains a network to reproduce the raw sensor signals and raises an alarm when the reproduction is poor; TimesNet [5], TranAD [6], and Anomaly Transformer [7] follow this recipe. The idea is sound: a network trained only on healthy data should copy healthy data well and faulty data badly. In practice, much of raw telemetry cannot be copied at all. Sensor jitter is random, so no network can reproduce it, and part of the error never goes away. Slow changes such as thermal drift or a switch in operating mode also degrade the copy even though nothing is broken. These errors are then added across 25–55 channels measured in different units, so a single noisy channel can dominate the total. The result is an error score with occasional large spikes on perfectly healthy data.

These spikes defeat the alarm threshold. We set each detector's threshold on a block of healthy data with Streaming Peaks-Over-Threshold (SPOT) [8], a standard method that models the extreme tail of the score and targets a low false-alarm rate. On later healthy data from the same streams, the reconstruction detectors nevertheless alarm at 22%–24% of time steps under causal reporting, roughly one step in four or five. At that rate, their recall says little about fault recognition. Faults last many steps, so a detector that ignores the data and alarms at random 24% of the time would still place at least one alarm inside 60.0% of fault events. TimesNet catches 58.8%: its recall comes from alarming often, not from recognizing faults.

Gap 2, one anomaly score allows a few drifting features to hide a fault: A Joint-Embedding Predictive Architecture (JEPA) [9], [10] avoids copying raw signals. An encoder compresses each window of sensor data into a short summary of 32 numbers, and a predictor learns to forecast the summary of the data that comes next. The summary keeps the structure of the signals and discards much of the unpredictable jitter, so its prediction errors are far better behaved and false alarms on healthy data fall by an order of magnitude or more (Contribution 3).

Standard JEPA detectors, however, still squash the 32 prediction errors into one number and compare it against one threshold. That threshold must sit above everything seen on healthy data. If one or two summary features drift slightly over time, as happens when a machine warms up or changes duty cycle, their errors grow and the shared threshold rises to accommodate them. A fault that shifts the other 30 features by a moderate amount then stays below the raised threshold and goes unnoticed. We call these drifting features *hijacker dials*. The cost is visible in streaming use: under causal reporting, single-score JEPA detectors reach a Point-F1 (alarm accuracy

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scored at every time step) of only 0.050–0.055 and take 138–154 steps to notice a fault.

Gap 3, reported accuracy often depends on how the test is scored: Two common evaluation shortcuts make detectors look better than they would be in service. Point adjustment counts a whole fault segment as detected if any single step inside it is flagged, which rewards frequent alarms [11]. Thresholds tuned with test labels, or calibrated on the same data that shaped the model, give optimistic false-alarm rates. We therefore score every time step without point adjustment, set all thresholds on healthy data only, keep model fitting, checkpoint selection, adaptation, and threshold calibration on separate blocks of data, and report final results on streams never inspected during design.

Our approach, one circuit breaker per dial: We replace the single alarm with a panel of circuit breakers, much like the breaker box in a building, where a short in one room trips one switch regardless of the total load. The predictor forecasts the summary of the next 16, 32, and 64 steps, and for each of the 32 summary features we track how large its error is at each horizon and whether that error leans steadily in one direction. Each such reading is a *dial* with its own threshold fitted on healthy data, and the system alarm fires when enough dials trip together (a quorum). Dials that wander on healthy data receive wider thresholds, so a hijacker widens only its own threshold and can no longer mask the rest of the panel. We target faults that appear suddenly or develop within about one context window (256 steps), and we study two versions (Fig. 1):

1) *Laminar-JEPA-Lite*, our preferred deployment configuration, predicts three horizons in parallel behind a single layer-normalization (LayerNorm) step [12] and watches $4 \times 32 = 128$ dials under one quorum. It has 35%–52% fewer parameters and runs $2.43\times$ faster.
2) *Laminar-JEPA*, the full reference variant, watches $7 \times 32 = 224$ dials covering four views of each coordinate's error, plus 32 fast onset relays and an optional weight-editing step that damps drifting coordinates.

A. Contributions

Design choices were tuned on 33 development streams. The final evidence comes from 118 untouched streams from 9 datasets (of 141 streams from 12 datasets) that played no role in design; checkpoint selection, model adaptation, and threshold calibration each use a separate block of healthy data (Section V-B). We report two alarm modes: *causal*, where each alarm uses only data available at that moment, and *buffered*, where a short look-back window supports after-the-fact attribution.

1) *More faults caught at low false-alarm rates:* On the 118 untouched streams, *Laminar-JEPA-Lite* catches 63.8% of fault events under causal reporting, against 52.5% for a single-score multi-horizon JEPA. This event recall (ER) gain holds on all 9 datasets (sign-flip $p = 0.0039$) and in all 12 leave-one-dataset-out (LODO) folds of the full archive. Causal Point-F1 rises from 0.076 to 0.079 ($p = 0.014$) and buffered Point-F1 from 0.143 to 0.156 ($p = 0.022$), and the panel beats simpler per-coordinate detectors tuned to the same overall alarm rate.
2) *Where the gains come from:* Controlled ablations on matched backbones show that LayerNorm compresses heavy error tails and that centering the summaries reduces drift, while spectral reweighting brings no measurable benefit.
3) *Far fewer false alarms than reconstruction:* On 19 development streams, latent prediction keeps the causal false-positive rate (FPR) at 0.6%–2.4%, against 22.2%–24.3% for reconstruction detectors. When every detector is allowed the same 1% test FPR, *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% and 56.0% of events, against 21.7% for TimesNet.
4) *Limits:* Proposition 1 bounds how concentrated the horizon errors can be, yet reweighting scores by this quantity does not help. Strong autocorrelation means that 300 effective calibration samples cannot pin down a 99.9th-percentile threshold precisely. An optional single-channel branch lifts recall of injected single-sensor faults to 80.7%–96.5% but raises nominal causal FPR from 1.5% to 6.8%–19.4%. Slow ramps lasting four or more context windows and natural sensor failures remain out of reach.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep detectors typically forecast or reconstruct raw windows and threshold $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$: autoregressive long short-term memory (LSTM) forecasters [4] model spacecraft channels directly; TranAD [6] pairs adversarial training with self-conditioning attention; Anomaly Transformer [7] contrasts Gaussian prior and series-level attention; and TimesNet [5] reshapes 1D series along Fast Fourier Transform (FFT) periods for 2D convolution. Equal channel-wise reconstruction losses fit stochastic jitter, so summing errors across $K \geq 25$ channels yields heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) spikes and high false alarms under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Representation methods encode windows into compact latents [14]. Contrastive approaches—temporal-contextual contrasting [15], TS2Vec [16], Neural Contextual Anomaly Detection (NCAD) [10] with synthetic anomalies, and dual-attention DCdetector [17]—depend on whether synthetic perturbations or patch discrepancies match test faults (Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] predict target embeddings z_{tgt} from context z_{ctx} without observation decoding or negative pairs, avoiding collapse via exponential moving average (EMA) targets [20] or variance-covariance terms [21]–[23].

Residual Decomposition, Conformal Thresholding, and Statistical Process Control: Telemetry reflects underlying system dynamics [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], Hamiltonian/Lagrangian nets [27], [28], and Koopman lifts [29]–[33] model structured dynamics, though partial differential equation (PDE) penalties require known laws and exhibit stiff gradients on uncalibrated sensors [34]; we instead borrow the

algebraic mean-plus-fluctuation identity of Reynolds averaging [35] strictly as a multi-horizon residual summary. Multivariate statistical process control (SPC) charts such as Hotelling's T^2 [36] and Multivariate Exponentially Weighted Moving Average (MEWMA) [37] collapse whitened coordinates into one scalar, as do 1D deep scores calibrated via extreme-value theory [38] and Generalized Pareto Distribution (GPD) tails in SPOT [8] under serial dependence [39]. Per-coordinate alternatives such as univariate Cumulative Sum (CUSUM) banks [40] and per-coordinate split-conformal rules under Bonferroni, Šidák, or Benjamini–Hochberg (BH) false-discovery-rate (FDR) control [41] hit a discrete calibration floor $(N_{\text{cal}}+1)^{-1}$ on short histories (Section VI-A). All methods are evaluated without point adjustment [11], [13], [42].

III. Multi-Sensor Telemetry Benchmark Suite

Each test case in this paper is a *stream*: one recording from one system, with K sensor channels sampled over time. Every stream has a healthy (nominal) history, used for training and calibration, followed by a labeled test segment that contains the faults. We use two collections of streams. A 45-stream development suite drawn from 11 datasets guided every design decision and supplies the main development benchmark. A larger 141-stream archive drawn from 12 datasets supplies the final evaluation, and 118 of its streams were never inspected during design.

Development suite: The 45 streams come from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [13], and mTSBench [45]. They cover three kinds of systems with channel counts from 2 to 72 and very different fault types:

1) *Spacecraft and solar telemetry:* six SMAP streams (K=25), six MSL rover streams (K=55), and SWAN-SF solar magnetograms [46] (K=38). NASA labels each SMAP and MSL fault as either an out-of-range *point* fault, where a value leaves its normal range, or an in-range *contextual* fault, where values stay in range but behave abnormally together [4] (Table V(a)).
2) *Wearable motion sensing:* six Daphnet Freezing-of-Gait (FoG) streams [43] recorded at 64 Hz (K=9, 72 episodes), so the S=64-step target window spans one second.
3) *Industrial processes and infrastructure:* 14 Gasoil Heating Loop (GHL) streams (K=16) whose 26 faults are slow ramps, the Genesis pick-and-place testbed (K=18), GECCO water chemistry (K=9), Metro I-94 traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72).

Thirty-three of these streams form the development benchmark: 19 non-GHL streams and the 14 GHL streams, spanning 7 datasets. Inclusion required enough healthy data for calibration ($N_{\text{cal}} \geq 500$ windows), at most 55 channels, and at most 1.2×10^5 test steps (Table IV(c)). The other 12 streams are reserved: five have very short healthy histories (764–1,492 steps; CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), the six CICIDS streams exceed the channel limit, and GECCO is evaluated in the archive instead.

Evaluation archive: The archive contains every qualifying stream from 12 datasets, requiring at least 100 calibration windows and at least one labeled anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, and 13 OPPORTUNITY streams, plus single streams from GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN. Healthy histories range from 1,920 to 132,481 steps and test segments from 2,158 to 114,001 steps. Removing the 23 streams used to tune the method leaves 118 untouched streams from 9 datasets (Section V-B).

How each stream is split: Every healthy history is divided in time order. The first 65% fits the model, and its first part also sets the per-channel standardization ($\boldsymbol{\sigma}$ floored at 10^{-2}). The next 17.5% calibrates alarm thresholds ($\mathbf{X}_{\text{cal}}$), and after a 64-step gap the last 17.5% measures false alarms on healthy data the thresholds have never seen ($\mathbf{X}_{\text{hold}}$). Archive runs further divide the model-fitting block (Section V-A).

Caveats: SMAP and MSL contain documented labeling flaws [42], and streams from the same archive often share hardware, so they are not fully independent. We therefore also treat whole datasets as units of analysis (7 for the development benchmark, 9 for the untouched streams, 12 for the full archive), using Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and leave-one-dataset-out audits (Section V-A).

IV. Methodology: Laminar-JEPA and Laminar-JEPA-Lite

Fig. 1 shows the pipeline from raw telemetry $\mathbf{X} \in \mathbb{R}^{T\times K}$ (K channels over T steps) to alarms. In plain terms, it has four stages: (1) encode each window of recent data into a short summary vector; (2) predict the summaries of the next few stretches of data; (3) turn the prediction errors into a set of *dials*, each measuring one aspect of the error; and (4) give every dial its own threshold and raise an alarm when enough dials trip together. The networks are trained on healthy data only; stabilization weights and alarm thresholds are then fitted on later healthy blocks and frozen.

A. Stage 1: Multiscale Encoder and Barycenter Centering

A sliding window of length $W = C + S = 320$ advances one step at a time. At each position t it splits into a *context window* $\mathbf{x}_{\text{ctx}}$ of the past $C = 256$ steps and a *target window* $\mathbf{x}_{\text{tgt}}$ of the next $S = 64$ steps, which ends at the decision time $\tau = t+S-1$. Telemetry mixes slow trends with fast vibration, so the encoder first separates time scales. Multichannel inputs ($K > 1$) pass through a causal per-channel convolution and are split into four smooth frequency bands (dyadic shells of a 256-point real fast Fourier transform (rFFT), forming an amplitude partition of unity). Each band has its own causal convolution (kernels $\{15, 9, 5, 3\}$), and a softmax router feeds the bands into a causal dilated temporal convolutional network (TCN) [48]. Single-channel inputs skip the band split.

The encoders follow the Joint-Embedding Predictive Architecture (JEPA) [9]: instead of reconstructing raw waveforms, an online encoder E_θ and a target encoder E_ϕ map context and target windows to $D = 32$-dimensional summaries, each

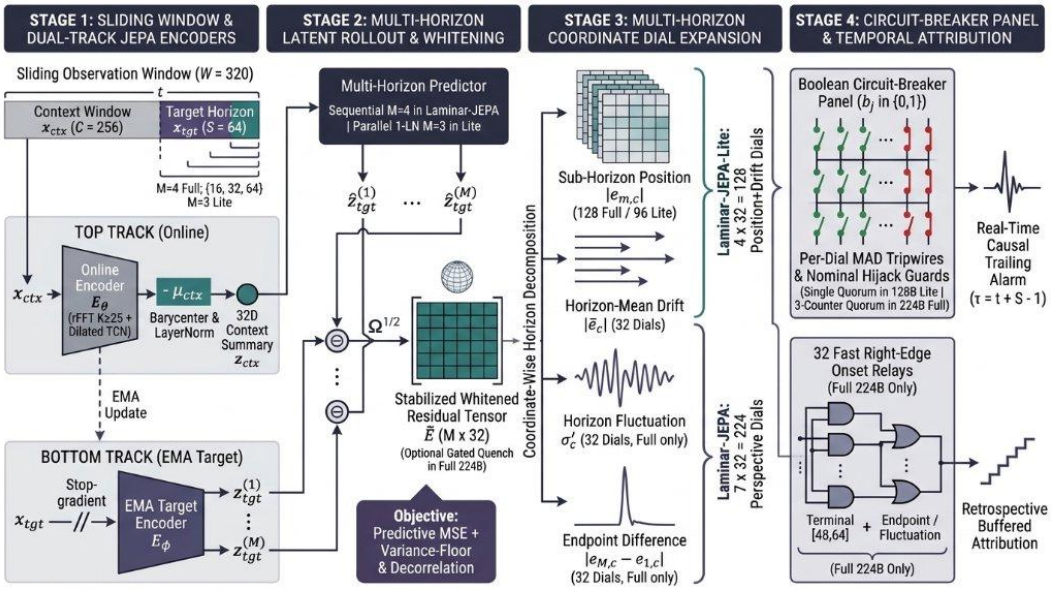


Fig. 1. Overview of the four-stage *Laminar-JEPA* and compact *Laminar-JEPA-Lite* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with conditional rFFT splitting ($K \geq 25$ in Lite), barycenter centering, and LayerNorm; **(2)** multi-horizon rollout (M=4 sequential in full *Laminar-JEPA*; M=3 parallel single-LayerNorm in *Lite*) under variance-floor and decorrelation loss plus resolvent whitening into $\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$; **(3)** horizon expansion into $7 \times 32 = 224$ dials (full) or $4 \times 32 = 128$ position-plus-drift dials (*Lite*); and **(4)** per-dial Boolean circuit breakers emitting causal (τ=t+S−1) or prefix-buffered alarms.

normalized by LayerNorm to length close to $\sqrt{D}$. The target encoder is a slowly updated copy of the online one, an exponential moving average (EMA) with momentum $m = 0.996$ (0.995 on archive cohorts) [18], [20]. Two problems arise in practice. Deep dilated TCNs blur brief spikes at fault onset, and changes in operating regime shift all summaries together. Two encoder components address them:

1) *Multi-depth tapping (keeps fast transients):* Each encoder also reads three intermediate feature maps ($d_{\text{model}} = 48$): the spectral stem, a middle TCN block, and the final TCN block. For each tap, the last time step and the root-mean-square (RMS) energy over time are projected to $\mathbb{R}^D$ by a two-layer GELU head and blended with the deep TCN output through a bounded gate ($g \leq 0.5$).
2) *Barycenter centering (removes common offsets):* During training, a running mean of the context summaries $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ (update rate 0.05, shared with E_ϕ) is subtracted from both context and target summaries before the final LayerNorm. Centering removes offsets learned in training; it does not guarantee invariance to unseen drift.

B. Stage 2: Multi-Horizon Prediction

The simplest JEPA detector, the single-horizon baseline *TS-JEPA*, uses a two-layer multi-layer perceptron (MLP) to predict one summary $\hat{z}_{\text{tgt}}$ of the whole 64-step target window, trained with predictive mean-squared error (MSE) and a VICReg-style log-determinant penalty ($\beta = 0.05$) [21]. Summarizing all 64 steps in one vector can average away a short fault. *Laminar-JEPA* therefore predicts how the target unfolds: it summarizes $M = 4$ nested prefixes of the target window, the first $S_m \in \{16, 32, 48, 64\}$ steps. Starting from z_{ctx} plus learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, a recurrent predictor P_{sub} produces $\hat{z}_{\text{tgt}}^{(1)}, \ldots, \hat{z}_{\text{tgt}}^{(M-1)}$ in sequence, and an anchor head P_θ predicts the full-window summary $\hat{z}_{\text{tgt}}^{(M)}$. The loss weights the full window by one half and splits the other half equally across the shorter prefixes (w_m), and adds a penalty $\boldsymbol{\Delta}_\gamma$ that keeps the summaries from collapsing to a constant:

$$\mathcal{L}_{\text{Lam}} = \sum_{m=1}^{M} \frac{w_m}{D} \left(\left\| \hat{z}_{\text{tgt}}^{(m)} - \text{sg}\left(z_{\text{tgt}}^{(m)}\right) \right\|_2^2 + \left\| \boldsymbol{\Delta}_\gamma\left(\hat{z}_{\text{tgt}}^{(m)}\right) \right\|_F^2 \right), \tag{1}$$

where $\text{sg}(\cdot)$ stops gradients through E_ϕ. Log-determinant penalties need batches larger than D; $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ works on small batches. Its diagonal entries $\sqrt{\max(0, \gamma - \sigma_d)}$ push every coordinate's standard deviation above $\gamma = 0.75$, and its off-diagonal entries $\Sigma_{dj}/\sqrt{2}$ penalize covariance between coordinates. At batch size $B = 16 < D = 32$, this raises the effective rank of the target summaries, $\text{RankMe}(z_{\text{tgt}})$ [49], from 11.6 to 19.8.

C. Stage 3: Residual Whitening and Dial Expansion

Stage 3 turns prediction errors into dials. The signed errors $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ for the $M = 4$ horizons form a 4×32 error matrix $\mathbf{E}(t)$. Its entries have different natural sizes and are correlated, so they are first *whitened*: rescaled to unit variance and decorrelated using statistics from healthy training windows. With the residual covariance $\hat{\boldsymbol{\Sigma}}_{\text{res}} = \mathbf{U}\,\text{diag}(\lambda_k)\mathbf{U}^\top$, we use the damped precision matrix $\boldsymbol{\Omega} = \mathbf{U}\,\text{diag}\big(\lambda_k/(\lambda_k + 10^{-3})^2\big)\mathbf{U}^\top$. Multiplying centered errors by $\boldsymbol{\Omega}^{1/2}$ gives whitened errors $e_{m,c}^{\text{white}}(t)$; the damping shrinks directions with almost no training variance instead of amplifying their noise.

1) *Stabilization on a separate healthy block $\mathbf{X}_q$:* A coordinate whose error grows between training and later healthy data would be inflated by whitening. We detect such coordinates on a separate healthy block $\mathbf{X}_q$, never on the block $\mathbf{X}_{\text{cal}}$ that later sets alarm thresholds (Section V-A), and blend them toward a simpler per-coordinate scaling $\hat{e}_{m,c}^{\text{raw}}(t)$. The

blend starts when the whitened RMS on $\mathbf{X}_q$ exceeds 1.10, and all weights are frozen before $\mathbf{X}_{\text{cal}}$ is used:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e^{\text{white}}_{m,c}(t) + (1 - w_{m,c})\, \hat{e}^{\text{raw}}_{m,c}(t), \qquad w_{m,c} = \frac{1}{1 + 2\max\big(0, \text{RMS}^{\text{white}}_q(m,c) - 1.10\big)^2}. \tag{2}$$

The full *Laminar-JEPA* can also run an optional gradient-free *weight-space quench*, which edits the few predictor outputs that drift most on healthy data and keeps the edit only if false alarms fall. It uses only $\mathbf{X}_{\text{fit}}$ and $\mathbf{X}_q$ (legacy fixed-epoch development runs without $\mathbf{X}_q$ use $\mathbf{X}_{\text{cal}}$; Table I). Coordinates are ranked by a median-absolute-deviation (MAD) standardized instability score that combines mean and scale shifts from $\mathbf{X}_{\text{fit}}$ to $\mathbf{X}_q$, drift between the halves of $\mathbf{X}_q$, and residual autocorrelation, with a negative term for isolated spikes. For the top $k \in \{4, 2\}$ coordinates, four steps run in order. *(Q1)* Damp the selected predictor output rows and LayerNorm gains and shift $\mathbf{p}_m$ toward the $\mathbf{X}_q$ means; through the shared 32D LayerNorm this also moves unselected coordinates (Section VI-A). *(Q2)* Move residual centers 75% of the way toward the post-edit $\mathbf{X}_q$ means. *(Q3)* Scale off-diagonal precision couplings by 0.15 and rescale whitened $\mathbf{X}_q$ RMS. *(Q4)* Keep the edit (preferring $k = 2$ on ties) only if the $\mathbf{X}_q$ half-split FPR improves while at least 92% of the pre-edit quorum thresholds are preserved; then freeze all parameters.

2) *Expansion into $7D = 224$ dials:* For each of the 32 coordinates, the four horizon errors describe how the prediction error evolves across the target window. Borrowing the mean-plus-fluctuation split of Reynolds averaging [35] purely as algebra, we split them into a horizon mean $\bar{e}_c(t) = M^{-1}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and a zero-mean fluctuation $F_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$. Four views of each coordinate give $7D = 224$ nonnegative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$: (a) $4D = 128$ *sub-horizon positions* $|\tilde{e}_{m,c}(t)|$, the error size at each horizon; (b) $D = 32$ *horizon-mean drifts* $|\bar{e}_c(t)|$, a persistent offset shared by all horizons; (c) $D = 32$ *horizon fluctuations* $d_{\text{fluc},c}(t) = \sqrt{M^{-1}\sum_{m=1}^{M} F^2_{m,c}(t)}$, how much the error changes across horizons; and (d) $D = 32$ *endpoint differences* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$, error that appears late in the window, as at a fault onset.

The proposition below states two exact properties of this split. Part (i) says that drift and fluctuation measure non-overlapping parts of the position energy, and that the endpoint dial cannot exceed a multiple of the fluctuation dial. Part (ii) says the fluctuation pattern lives in at most $M - 1$ directions and defines R_{LT}, which is 1 when one pattern dominates and $1/(M-1)$ when the energy is spread evenly. These are geometric facts about the representation, not accuracy guarantees.

Proposition 1 (Horizon Energy Split and Spectral Concentration Bounds)**.** *For $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) with horizon mean $\bar{\mathbf{e}}(t)$ and fluctuation matrix $\mathbf{F}(t) = \tilde{\mathbf{E}}(t) - \mathbf{1}_M \bar{\mathbf{e}}(t)^\top \neq \mathbf{0}$: (i) for every coordinate c, $M^{-1}\sum_{m=1}^{M} |\tilde{e}_{m,c}(t)|^2 = |\bar{e}_c(t)|^2 + d^2_{\text{fluc},c}(t)$ and $d_{\text{accel},c}(t) \leq \sqrt{2M}\, d_{\text{fluc},c}(t)$; and (ii) the $M \times M$ horizon Gram matrix $\mathbf{G}(t) = \mathbf{F}(t)\mathbf{F}(t)^\top \in \mathcal{S}^M_+$ has ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ whose scale-invariant dominant-mode share for any $p > 0$ [50] satisfies*

$$R_{LT}\big(\mathbf{F}(t)\big) = \frac{\lambda^p_M}{\sum_{m=1}^{M} \lambda^p_m} \in \Big[\frac{1}{M-1}, 1\Big], \tag{3}$$

reaching $\frac{1}{M-1}$ iff energy is isotropic across all $M-1$ active modes ($\lambda_2 = \cdots = \lambda_M > 0$, requiring $D \geq M-1$) and 1 iff $\mathbf{F}(t)$ has rank one. Empirically, scalar reweighting by R_{LT} has a null effect (Section VI).

Proof. Zero column sums ($\sum_{m=1}^{M} F_{m,c} = 0$) cancel the cross term in $M^{-1}\sum_{m=1}^{M}(\bar{e}_c + F_{m,c})^2$, and $(F_{M,c} - F_{1,c})^2 \leq 2(F^2_{M,c} + F^2_{1,c}) \leq 2M\, d^2_{\text{fluc},c}$ gives (i). For (ii), $\mathbf{F}^\top \mathbf{1}_M = \mathbf{0}_D$ implies $\mathbf{G}\mathbf{1}_M = \mathbf{0}_M$ ($\lambda_1 = 0$), while $\|\mathbf{F}\|^2_F > 0$ implies $\lambda_M > 0$. Summing $0 \leq \lambda^p_m \leq \lambda^p_M$ over $m \in \{2, \ldots, M\}$ yields $\lambda^p_M \leq \sum_{m=2}^{M} \lambda^p_m \leq (M-1)\lambda^p_M$; inverting and multiplying by λ^p_M gives (3) and its equality conditions, and nonzero scalar rescaling cancels. □

D. Stage 4: Circuit-Breaker Panel

Instead of adding all 224 dials into one score, Stage 4 (Algorithm 1) gives each dial its own threshold, learned from the healthy calibration block. A dial *trips* when it exceeds its threshold, and an alarm fires when enough dials trip at once. Because all dials come from the same 4×32 error matrix, their trips are correlated (median effective rank 65 of 224 on the development cohort; the variance of the trip count is $7.2\times$ that of independent dials). We therefore do not assume independence; the alarm quorums are set from the trip counts actually observed on healthy calibration data:

1) *Per-dial thresholds and hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ is standardized by its calibration median m_j and scaled MAD $s_j = 1.4826\ \text{MAD}(d_{j,\text{cal}}) + 10^{-4}$, giving a nonnegative score $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. Its tripwire τ_j is the 98.5th percentile of z_j on calibration data. Two hijack guards widen the tripwires of suspect dials: $\gamma^{\text{cal}}_j \geq 1$ widens a dial whose calibration tail grows over time (second-half 98th percentile above $1.35\times$ the first half), and $\gamma^{\text{sc}}_j \geq 1$ widens a dial much noisier than its peers (scale above $1.875\times$ its perspective-group median):
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\, \gamma^{\text{cal}}_j\, \gamma^{\text{sc}}_j. \tag{4}$$
2) *32 onset relays:* When a sudden fault first enters the right edge of the target window, the short prefixes ($m \in \{1, 2\}$) still cover healthy data, so few position dials trip. To catch this moment, each coordinate c has stricter 99.3rd-percentile edge tripwires ($\gamma^{1.25}$ guard) and fires a relay $r_c(t) \in \{0, 1\}$ when its full-window position dial ($m = 4$) trips together with its endpoint, fluctuation, or $m = 3$ dial.
3) *Alarm timing and quorum voting:* Under *causal attribution*, the trips from window $[t, t+S)$ are reported at its right end $\tau = t + S - 1$, the earliest moment all of its data exist; this fixes a reporting lag of $S - 1$ steps, separate from the detection delay $\Delta t = t_{\text{alm}} - t^*$ after fault onset t^*. Under *buffered attribution*, used for after-the-fact review, each trip is assigned to the steps it describes: position trips to their prefix $[t, t+S_m)$, drift and fluctuation trips to $[t, t+S)$, and endpoint trips to $[t+S_3, t+S)$. Counting active trips at each step gives $N_{224}(\tau)$, the position sub-count $N_{128}(\tau) \subset N_{224}(\tau)$, and the relay

count $N_{\text{edge}}(\tau)$. For ranking metrics, the continuous score $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ counts position trips and relays twice (fixed on development streams before held-out testing; Section VI-A). The binary alarm fires when any count exceeds its quorum:

$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \ \vee \ N_{128}(\tau) > K_{128} \ \vee \ N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \tag{5}$$

with quorums set on calibration counts: $K_{224} = \max(1.12\, q_{0.995}, 5.0)$, $K_{128} = \max(1.12\, q_{0.995}, 3.5)$, and $K_{\text{edge}} = \max(1.25\, q_{0.999}, 2.5)$.

E. *Compact Variant:* Laminar-JEPA-Lite

Laminar-JEPA-Lite keeps the same four stages but makes three simplifications, each supported by the ablations in Section VI. Together they cut trainable parameters by 35%–52% (141.9k at $K < 25$ and 222.3k at $K \geq 25$, vs. 295.8k), speed up windowed GPU inference by 2.43× (3.14× at $K < 25$), and need no weight editing:

1) *Smaller encoder:* A fifth dilated block blurred short-lived anomalies without making the summaries richer, so *Laminar-JEPA-Lite* uses four TCN blocks (dilations $\{1, 2, 4, 8\}$) and 64D pooling heads. The frequency-band front-end is used only on high-channel streams ($K \geq 25$), where it helped; smaller streams use a direct causal projection.
2) *Parallel three-horizon predictor:* One residual two-layer GELU MLP predicts $M = 3$ prefixes ($S_m \in \{16, 32, 64\}$) in parallel from $z_{\text{ctx}} + \mathbf{p}_m$, with a single final LayerNorm. This removes the lag of sequential prediction and the coupling between stacked normalizations, and shrinks $\mathbf{\Omega}$ to 96×96.
3) *128-dial panel with one quorum:* Only the 96 position dials $|\tilde{e}_{m,c}(t)|$ and the 32 drift dials $|\bar{e}_c(t)|$ are kept, each calibrated by (4). Fluctuation and endpoint dials, onset relays, and quenching are removed. Steps are ranked by the plain count $N_{128}(\tau)$, and the alarm uses one quorum, $K_{\text{Lite}} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\ 3.5\big)$.

1D single-score controls: To measure what the breaker panel adds over a single score on the same backbone (Section VI), the 1D controls pool a residual vector into one number: an equal blend of its ℓ_2 norm and a softmax-weighted coordinate RMS ($\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA pools $\mathbf{e}(t) \in \mathbb{R}^{32}$; 1D Multi-Horizon JEPA combines the pooled full-tensor and peak-horizon scores by root-mean-square (with optional R_{LT} modulation $\alpha_{\text{LT}} \in \{0.12, 0\}$). Both are thresholded with SPOT [8], [38], [51].

V. Experimental Protocol and Evaluation Metrics

A. Setup and Calibration Protocol

All architectures use context $C = 256$, target horizon $S = 64$ ($W = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), latent dimension $D = 32$, nominal training stride 16, and evaluation stride 1. Models are trained with AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) and zero per-dataset tuning: batch 16 for 15–20 epochs on development cohorts, and batch 16–64 for 10–12 epochs with gradient clipping (1.0) and cosine annealing on archive cohorts.

Algorithm 1 Laminar-JEPA (224-Bit) and Laminar-JEPA-Lite (128-Bit)

Require: Nominal history $\mathbf{X}_{\text{nom}}$, test stream $\mathbf{X}_{\text{test}}$, tier $\in$ {Full, Lite}, attribution mode $\in$ {causal, buffered}.
Ensure: Per-timestep binary alarms $\hat{y}(\tau) \in \{0, 1\}$ and continuous score $N_{\text{tot}}(\tau)$.

Phase 1: Disjoint Training, Whitening & Adaptation ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q$)
Split $\mathbf{X}_{\text{nom}}$ chronologically into S-isolated ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q, \mathbf{X}_{\text{cal}}, \mathbf{X}_{\text{hold}}$) and standardize channels using $\mathbf{X}_{\text{fit}}$ alone.
Set nested prefixes $S_{1:4} = \{16, 32, 48, 64\}$ (Full) or $S_{1:3} = \{16, 32, 64\}$ with $K \geq 25$ rFFT gate (Lite).
for stride-16 training batches in $\mathbf{X}_{\text{fit}}$ **do**
  Encode $z_{\text{ctx}} = E_\theta(\mathbf{x}_{\text{ctx}})$ and targets $z_{\text{tgt}}^{(m)} = E_\phi(\mathbf{x}_{\text{tgt}}^{(m)})$ with barycenter shift $-\bar{\boldsymbol{\mu}}_{\text{ctx}}$.
  Roll out sequential $\{P_{\text{sub}}, P_\theta\}$ (Full) or parallel single-LayerNorm predictor (Lite); update θ via (1) and $\phi \leftarrow m\phi + (1-m)\theta$.
end for
Select checkpoint on $\mathbf{X}_{\text{sel}}$ (or fixed epochs) and fit $(\bar{\boldsymbol{\mu}}_{\text{res}}, \mathbf{\Omega} \in \mathcal{S}_+^{MD})$ on $\mathbf{X}_{\text{fit}}$.
if tier = Full and $q_{\text{wsp}} = 1$ **then**
  Rank drifting coordinates on $(\mathbf{X}_{\text{fit}}, \mathbf{X}_q)$; run Steps Q1–Q4 for $k \in \{4, 2\}$ and keep edit iff $\mathbf{X}_q$ split and 0.92-quorum gates pass.
end if
Fit and freeze stabilization scales and blending weights $w_{m,c}$ (2) on $\mathbf{X}_q$.

Phase 2: Untouched Circuit-Breaker Calibration on $\mathbf{X}_{\text{cal}}$
Compute frozen stabilized residuals $\tilde{\mathbf{E}}_{\text{cal}}$ on untouched $\mathbf{X}_{\text{cal}}$ via (2).
if tier = Full **then**
  Expand $\tilde{\mathbf{E}}_{\text{cal}}$ into $7D = 224$ dials (position, drift, fluctuation, endpoint) and 32 right-edge onset relays.
  Fit MAD scales (m_j, s_j), hijack guards γ_j, tripwires τ_j (4), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (5).
else ▷ Preferred Deployment Tier: Laminar-JEPA-Lite
  Extract $4D = 128$ position-plus-drift dials $(|\tilde{e}_{m,c}|, |\bar{e}_c|)$, fit τ_j (4), and set single quorum $K_{\text{Lite}} = \max(1.12\, q_{0.995}, 3.5)$.
end if

Phase 3: Online Inference on $\mathbf{X}_{\text{hold}}$ or $\mathbf{X}_{\text{test}}$
for each stride-1 window $[t - C,\ t + S)$ **do**
  Compute $\tilde{\mathbf{E}}(t) \to \mathbf{d}(t)$ and binary trips $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$ (plus relays $r_c(t)$ if Full).
  Attribute active counts to right edge $\tau = t + S - 1$ (causal) or prefix spans $[t, t + S_m)$ (buffered).
  Emit $\hat{y}(\tau) \in \{0, 1\}$ when active counts exceed $(K_{224}, K_{128}, K_{\text{edge}})$ (5) or K_{Lite}.
end for

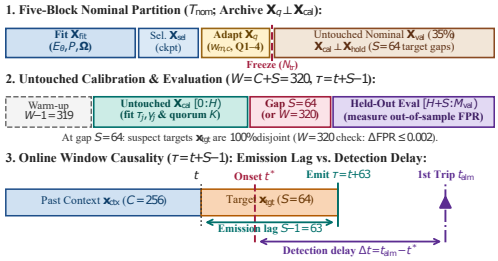


Fig. 2. Disjoint nominal evaluation protocol (C=256, S=64, W=320): (Row 1) five-block archive partition into S-isolated weight-fitting ($\mathbf{X}_{\text{fit}}$), checkpoint-selection ($\mathbf{X}_{\text{sel}}$), and nominal adaptation ($\mathbf{X}_q$) prefixes versus $\mathbf{X}_{\text{val}}$, (Row 2) untouched threshold calibration ($\mathbf{X}_{\text{cal}}$) and held-out nominal evaluation ($\mathbf{X}_{\text{hold}}$), and (Row 3) structural emission lag ($S-1$) versus empirical detection delay (Δt).

a) Disjoint Calibration Protocol: Each stream's nominal history $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Table I) into a model-fitting prefix $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = T_{\text{nom}} - \lfloor 0.35\, T_{\text{nom}} \rfloor$) and a held-out nominal suffix $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$: (1) *Stage A (Disjoint weight fitting, checkpoint selection, and nominal adaptation):* Reusing calibration windows for early stopping, blending weights (2), or weight quenching can make nominal thresholds optimistic.

TABLE I
NOMINAL PARTITIONING AND CALIBRATION PROTOCOL BY COHORT

Cohort (Tables)	Weights (%)	Ckpt. Sel.	Adapt. / Quench	Thresholds
Dev. 19 Cache (IIa,b, III)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	None (unblended)	$\mathbf{X}_{\text{cal}}$
Dev. 33 Suite (IIc, IV, V)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	Shared $\mathbf{X}_{\text{cal}}$	$\mathbf{X}_{\text{cal}}$
Dev. 19 4-Block (Sec. V-B)	$\mathbf{X}_{\text{fit}}$ (50)	Fixed ep.	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Archive 118/141 (VI)	$\mathbf{X}_{\text{fit}}$ (43)	$\mathbf{X}_{\text{sel}}$	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Held-Out / Test ($\mathbf{X}_{\text{hold}}$, $\mathbf{X}_{\text{test}}$)	Frozen	Frozen	Frozen	Frozen

All archive benchmarks (Table VI) therefore use a disjoint *five-block protocol* ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$) inside $\mathbf{X}_{\text{tr}}$: $\mathbf{X}_{\text{fit}}$ ($\approx 43\%$ of T_{nom}) fits channel standardization, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and precision $\boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$; $\mathbf{X}_{\text{sel}}$ ($N_{\text{sel}} \geq 96$ windows) selects the early-stopping checkpoint; and $\mathbf{X}_q$ ($N_q \in [160, 2{,}048]$ stride-1 windows separated from $\mathbf{X}_{\text{sel}}$ and $\mathbf{X}_{\text{val}}$ by $S = 64$-step target-isolation gaps) fits and freezes (2) and optional split-gated quenching (Steps Q1–Q4) before $\mathbf{X}_{\text{cal}}$ is ever touched. Fixed-epoch development suites (Table I) omit $\mathbf{X}_{\text{sel}}$, and the 19-stream residual cache also skips (2) and quenching. A four-block check ($\mathbf{X}_{\text{fit}} \to \mathbf{X}_q \to \mathbf{X}_{\text{cal}} \to \mathbf{X}_{\text{hold}}$) on the 19 development streams gives causal FPR_{hold} of 0.96% for *Laminar-JEPA-Lite* and 1.05% for *Laminar-JEPA*, matching Table IV, although the smaller fitting block (50% instead of 65%) lowers causal F1 ($0.133 \to 0.071$ and $0.144 \to 0.076$). On short histories, *two-fold chronological cross-fitting* swaps the two halves of $\mathbf{X}_{\text{val}}$ and alarms only when both folds agree ($\hat{y}^{(1)} \wedge \hat{y}^{(2)}$; Section VII-E). (2) *Stage B (Untouched circuit-breaker calibration and boundary isolation on $\mathbf{X}_{val}$):* Stride-1 windows of length $W = 320$ inside $\mathbf{X}_{\text{val}}$ yield $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window indices. Splitting at $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the untouched *nominal calibration block* $\mathbf{X}_{\text{cal}} = [0 : H)$ and *held-out nominal evaluation block* $\mathbf{X}_{\text{hold}} = [H+S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H+S)$ so calibration and evaluation share zero target observations (a full $G = 320$ window gap changes causal FPR by < 0.25 percentage points). On $\mathbf{X}_{\text{cal}}$ alone, the frozen model fits dial medians m_j, MAD scales s_j, hijack guards γ_j, tripwires τ_j (4), and quorums (K_{Lite} or $(K_{224}, K_{128}, K_{\text{edge}})$ in (5); τ_{SPOT} for 1D baselines), frozen on $\mathbf{X}_{\text{hold}}$ and $\mathbf{X}_{\text{test}}$. Raw window count $N_{\text{cal}} = H$ is distinguished from autocorrelation-adjusted *effective sample size* $N_{\text{eff},s} = N_{\text{cal}}/\tau_{\text{int},s}$ via stream- and statistic-specific Sokal initial-positive-sequence estimators (τ_{int} up to 206; Section VII-E).

b) Statistical Inference: We report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Dataset-clustered and stream-level inference: Dataset-clustered inference* (95% cluster-bootstrap CIs and exact 2^D dataset sign-flip permutation tests) is emphasized alongside stream-level paired Wilcoxon tests and 95% stream-bootstrap CIs (2,000 resamples). For the four focal causal contrasts on the 118 untouched streams (*Laminar-JEPA-Lite* vs. Multi-Horizon JEPA (1D) and TS-JEPA, on Point-F1 and ER), Benjamini–Hochberg (BH) [41] adjusted p-values are 0.014 and 3.6×10^{-4} (stream-level Point-F1) and 0.0078 and 0.012 (dataset sign-flip ER, positive on $9/9$ and $8/9$ datasets); other quoted p-values are raw and descriptive, and non-Exathlon subsets ($n = 101$, 8 datasets) verify sensitivity to archive composition. (2) *12-fold LODO audit:* Across all 12 LODO folds of the 141-stream archive, *Laminar-JEPA-Lite* keeps a positive gain on causal ER ($+0.099$ to $+0.141$ vs. Multi-Horizon 1D), buffered Point-F1, and causal Point-F1 vs. TS-JEPA ($\max p = 0.0054$); its causal Point-F1 gain over Multi-Horizon 1D holds in $9/12$ folds.

B. Cohorts and Baselines

Laminar-JEPA (w/o Quench) denotes the full variant without Steps Q1–Q4. Causal and buffered attribution are paired on the same splits within each cohort (aside from a single-horizon 130-stream buffered baseline screen in Section VII-A): (1) *Development cohorts (design-inspected, Table I):* The 19-stream residual cache (6 datasets) evaluates the dial ladder, component ablations, and conformal/statistical-process-control (SPC) regimes on the same unquenched *Laminar-JEPA* residuals (Table II(a,b), Table III). The 33-stream development benchmark (7 datasets: 19 non-GHL streams in Table II(c) plus 14 GHL streams in Table IV) compares both tiers against 1D JEPAs and observation-space baselines. Tripwire and quenching constants were fixed on 23 tuning streams (19 non-GHL plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`). (2) *Untouched held-out cohorts (primary generalization evidence):* Excluding those 23 design streams from the 141-stream archive (12 datasets, five-block protocol) leaves 118 untouched held-out streams across 9 datasets (101 non-Exathlon streams across 8 datasets; Table VI), supplemented by a single-seed *Laminar-JEPA* Q1–Q4 step-isolation audit on the same 118 streams (Section VI-A). None of the 118 held-out streams was inspected during architecture or threshold design.

Standardized Baseline Fairness Protocol: Scalar baselines (TS-JEPA, Multi-Horizon JEPA (1D), TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [17]) share the same chronological splits, training-only standardization, isolation gaps, and SPOT calibration ($u = p_{98}$, $q = 10^{-3}$), whereas conformal, SPC, and simpler coordinate detector-bank baselines in Tables III and VI share the same multi-horizon whitened residuals and nominal calibration alert rate. External neural baselines use 206k–649k trainable parameters; circuit-breaker panels add none. Runs used PyTorch 2.4 / CUDA 12.4 on an RTX 5090 (batched inference 0.15 ms/window for *Laminar-JEPA-Lite* and 0.38 ms for *Laminar-JEPA*; 188 MB peak memory); code: https://github.com/quaid281/NCAD.

C. Metrics

To prevent point-adjustment inflation [11] and aggregation ambiguity, all binary false-alarm and Point-F1 metrics are computed *per valid timestep* τ (never point-adjusted and never per-window without timeline attribution): FPR_{hold} averages $\hat{y}(\tau)$ over valid timesteps in $[H+S : M_{\text{val}})$, FPR_{test} averages $\hat{y}(\tau)$ over nominal test timesteps ($y(\tau) = 0$), and Point-F1, Precision, Recall, and precision-recall (PR) AUC (from $N_{\text{tot}}(\tau)$) score all valid test timesteps against $y(\tau) \in \{0, 1\}$. Event-level ER measures the fraction of contiguous $y = 1$ segments with ≥ 1 alarm, and delay $\Delta t = t_{\text{alm}} - t^*$ is the step lag from segment onset t^* to first in-segment alarm. Headline numbers are stream-level means of within-stream

seed averages unless labeled *median* or *Dataset-Macro*; we add stream medians for skewed FPR and use post-hoc test-FPR controls ($f^* \in \{0.01, 0.05\}$) for diagnosis only.

VI. Ablation Studies and Analysis

A. Ablation 1: Panel Components and Quenching

Table II(a),(b) and Fig. 3(a) evaluate perspective expansion and panel components on the 19-stream residual cache (not paired to the separately trained run in Panel (c)):

1) *Reference 224-dial panel vs. reduced position-plus-drift subsets:* We keep the 224-dial *Laminar-JEPA* as a reference because scoring all four views on one residual tensor shows which views help causal versus buffered reporting. Against 1D scoring, the 224 dials raise causal Point-F1 from 0.013 to 0.071 ($p = 0.0023$; buffered $+0.033$, $p = 0.53$). Pruning the fluctuation, endpoint, and relay dials to a 160-dial position-plus-drift subset helps causal reporting: Point-F1 rises to 0.098, ER to 59.1%, delay falls from 90.1 to 74.8 steps, and held-out FPR drops from 0.76% to 0.55% with 29% fewer dials. Buffered Point-F1 still favors the full panel (0.224 vs. 0.198). This result motivates *Laminar-JEPA-Lite*.
2) *Relays, PR-AUC score weighting, quorum rules, and hijack guards:* Removing the 32 onset relays lowers causal Point-F1 ($0.071 \rightarrow 0.068$), and replacing the three-counter quorum (5) with one 224-dial quorum lowers causal Point-F1 to 0.063 ($\text{FPR}_{\text{hold}} = 0.0055$; Table III). Replacing the double-position continuous score $N_{\text{tot}} = N_{224} + N_{128} + 2N_{\text{edge}}$ with unweighted $N_{224}+N_{\text{edge}}$ or N_{224} leaves binary alarms unchanged and changes PR-AUC by at most 0.004 on this cache and 0.0016 on the 118 untouched streams (*Laminar-JEPA-Lite* uses unweighted N_{128}). Removing hijack guards ($\gamma_j = 1$) raises causal Point-F1 to 0.073 (70.1-step delay) on this cache when nominal drift is mild, so the guards are a safeguard against drift rather than a source of accuracy. No panel component removes the $S-1$ reporting lag.
3) *Quenching step isolation and LayerNorm coupling:* We tested each quench step in isolation on single-seed *Laminar-JEPA* checkpoints across all 118 untouched streams (unquenched: causal $\text{FPR}_{\text{hold}} = 6.76\%$, F1 $= 0.0901$). Damping $k = 4$ predictor rows (Step Q1) shifts unselected coordinates by a median 11.1% of nominal RMS through 32D LayerNorm coupling and does not lower FPR_{hold} (7.06%). Recentering and decoupling (Steps Q2–Q3) lower it to 6.71% but cut causal F1 to 0.0826. The split gate (Step Q4, accepted on $39/118$ streams) reaches 6.62% with F1 unchanged (0.0904). Because ungated edits hurt $k = 2$ causal F1 on $11/19$ development streams ($0.141 \rightarrow 0.049$) and $k \in \{8, 12\}$ cuts PR-AUC ($0.292 \rightarrow 0.223$), *Laminar-JEPA-Lite* omits weight editing altogether.

Marginal, multiplicity-controlled, and system-matched detector banks: Simple per-coordinate alarms either fire constantly without a quorum, almost never fire under multiplicity correction on short calibration sets, or lose their calibration out of sample. We compare three calibration regimes on the same 19-stream whitened residuals without test-label tuning (Table III): (1) *Per-coordinate marginal union:* Thresholding 128 coordinates individually at $q_{0.985}$ or even at the panel's 0.34% rate without a quorum compounds false alarms across coordinates ($\text{FPR}_{\text{hold}} = 66.5\%$ and 40.6%). (2) *Family-wise and false discovery rate (FDR) multiplicity control:* Applying Šidák (0.5%) across 128 coordinates or BH FDR ($q = 0.005$) across 224 dials hits the discrete conformal p-value floor $(N_{\text{cal}} + 1)^{-1} > 3.9 \times 10^{-5}$, paralyzing causal recall (F1 ≤ 0.001). (3) *System-level matched nominal alarm rate:* Calibrating the aggregated statistic directly to each stream's nominal panel rate (0.34%) shows that discrete conformal $\min_j p_j$ still jumps to $\text{FPR}_{\text{hold}} = 21.1\%$ out of sample, because ties at the boundary fire whenever any coordinate exceeds its calibration maximum. Recalibrated Mahalanobis conformal, Hotelling T^2, MEWMA, and CUSUM charts inflate to 27.9%–58.6%. Among stable system-matched rules, the 160-dial position-plus-drift panel (F1 $= 0.098$, $\text{FPR}_{\text{hold}} = 0.55\%$) outperforms 128-coordinate top-k voting (0.068), continuous $\max_j z_j$ (0.056), and ℓ_2^2 conformal (0.072); Fisher/Stouffer pooling reaches similar F1 ($0.109/0.095$) at higher FPR_{hold} ($1.05\%/0.87\%$).

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

A frozen or flatlined sensor barely moves the latent summary: terminal LayerNorm spreads the small change across all 32 coordinates, so on `SMAP P-3` a flatline trips 12.5 dials versus 62.1 for a variance-matched harmonic flip, and a single faulty sensor is further diluted by the $K-1$ healthy channels. Extending Fig. 3(b) (n=6), we evaluate a causal channel branch across all 19 multivariate development streams ($K \in [5, 55]$) without per-stream or test-label tuning. Per channel, three 64-step rolling statistics—log-variance drop $-\ln(\text{Var}_{64}(x_c) + 10^{-4})$, training-fit first-order autoregressive (AR(1)) residual RMS, and mean shift $|\bar{x}_{64,c} - \mu_c|/\sigma_c$—are thresholded at $1.25\, q_{0.999}$ on nominal validation windows (floor 6.0 MAD, $1.25\times$ half-split drift multiplier) with a 24-of-40-step persistence gate. On the highest-variance active channel $c^\star = \arg\max_{c:\sigma_c^{\text{tr}} > 10^{-8}} \sigma_c^{\text{tr}}$, we place three nonoverlapping 200-step faults per stream inside clean test segments (57 events per class). For $s \leq t < s + 200$ using training $(\mu_c, \sigma_c, q_{.995,c})$:

$$x'_{t,c^\star} = \begin{cases} \mu_{c^\star} & \text{variance collapse,} \\ \mu_{c^\star} + 1.35(-1)^{t-s}\sigma_{c^\star} & \text{prediction shock,} \\ x_{s-1,c^\star} & \text{sensor freezing,} \\ \max(q_{.995,c^\star}, \mu_{c^\star} + 2.2\sigma_{c^\star}) & \text{saturation,} \\ x_{t,c^\star} + 2.5\sigma_{c^\star}(t-s)/199 & \text{gradual degradation,} \\ x_{t,c^\star} + 1.5\sigma_{c^\star} & \text{isolated bias.} \end{cases} \tag{6}$$

Outside $[s, s + 200)$, all channels remain unchanged. The variance-collapse union lifts injected-event recall for variance collapse, freezing, and saturation from 63.2%–82.5% to 93.0%–94.7%, and the full channel branch lifts recall on shocks, gradual degradation, and bias from 59.6%–75.4% to 80.7%–96.5%. However, on the *unmodified* test streams, causal FPR_{test} rises from 1.5% (standalone panel, F1 $= 0.071$) to 6.8% (variance-collapse union, F1 $= 0.070$) and 19.4% (full branch, F1 $= 0.129$). Synthetic single-channel injections do not establish detection on natural sensor failures, and the

TABLE II
CIRCUIT-BREAKER DIAL EXPANSION, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Horizon Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Horizon Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (2). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Laminar-JEPA (224-Bit)	0.92	0.190	0.013	0.128	0.144	**0.244**	**0.311**	**0.407**	0.673	0.795	0.612	0.757
Laminar-JEPA (w/o Quench)	0.88	0.190	0.016	0.128	0.142	0.235	0.304	0.381	**0.695**	0.815	**0.631**	0.765
Laminar-JEPA-Lite (128-Bit)	2.33	0.182	0.010	0.061	0.133	0.242	0.301	0.397	0.650	0.811	0.560	**0.772**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Multi-Horizon JEPA (1D)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Multi-Horizon JEPA (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

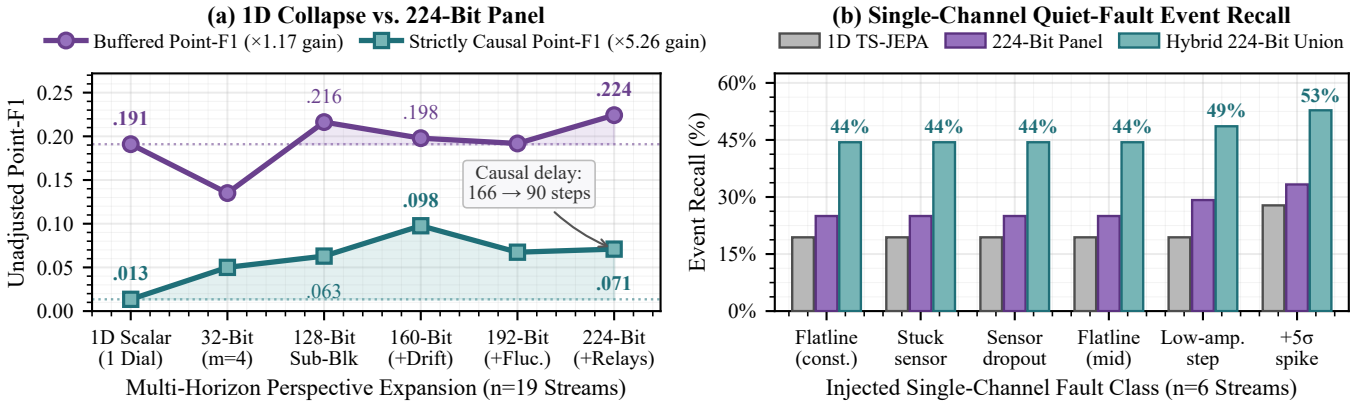


Fig. 3. Development studies. **(a)** Causal and buffered Point-F1 across the six-step expansion on cached whitened residuals from 19 streams (three seeds; no stabilization blend). **(b)** Event recall for injected single-channel faults in a separate six-stream study, comparing 1D TS-JEPA, 224-Bit Breaker, and their channel-level union.

full union's 19.4% false-alarm rate makes it unsuitable as a default detector.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining the multi-horizon JEPA backbone with and without 4-shell rFFT splitting on five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, `GECCO`; three seeds) shows a channel-count threshold: shell splitting improves high-channel spacecraft detection (K=55 `MSL M-1` buffered F1 0.085 → 0.410; K=25 `SMAP P-3` causal F1 0.212 → 0.321), but lowers F1 on both K=9 streams (`Daphnet`, `GECCO`), setting the $K \geq 25$ spectral gate in *Laminar-JEPA-Lite*.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 16 matched-budget arms in Table II(c) isolates five mechanisms: (1) raw-observation forecasting and autoencoding share heavy tails ($\gamma_2 \approx 6.2$), large drift ($D_{\text{KS}} \approx 0.45$), and ≈ 25% causal FPR; (2) a D=32 LayerNorm bottleneck in observation space cuts kurtosis ($\gamma_2 = 0.91$) and halves FPR, yet raw targets keep drift high ($D_{\text{KS}} = 0.376$); (3) learned D=32 latent targets and whitening halve drift ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179–0.190 across multi-horizon JEPA tiers, $p \leq 0.001$); (4) linear PCA and Isolation Forest keep low causal FPR (0.021 and 0.039) but reach only 0.017 and 0.041 causal Point-F1; and (5) frozen targets and ℓ_2 scoring retain < 2% causal FPR, while latent context autoencoding lowers causal Point-F1 (0.112 vs. 0.161 for TS-JEPA).

TABLE III
COORDINATE DETECTOR-BANK REGIMES ON CACHED RESIDUALS

Detector & Calibration Regime	Pt-F1 ↑	ER ↑	$\mathrm{FPR}_{\mathrm{hold}}$ ↓	$\mathrm{FPR}_{\mathrm{test}}$ ↓
Multi-horizon circuit-breaker panels (nominal quorum)				
160-dial pos.+drift (single quorum)	**0.098**	0.591	0.0055	0.0087
224-dial panel (deployed)	0.071	0.564	0.0076	0.0150
224-dial single quorum (no relay)	0.063	0.564	0.0055	0.0088
1. Per-coordinate marginal union (no quorum, 128 coords)				
Per-coord. $q_{.985}$ union	0.291	0.913	0.6651	0.7353
Per-coord. .0034 marginal union	0.303	0.874	0.4057	0.4765
2. Family-wise / FDR multiplicity control (α, q=.005)				
Per-coord. conformal (Šidák)	0.000	0.000	**0.0000**	**0.0000**
224-dial conformal BH FDR	0.000	0.026	**0.0000**	**0.0000**
3. System-level matched nominal rate (mean .0034 on $\mathbf{X}_{cal}$)				
Raw 128-coord. top-k vote	0.068	0.564	0.0052	0.0096
Continuous 128-coord. $\max_j z_j$	0.056	0.658	0.0078	0.0147
Discrete conformal $\min_j p_j$	0.265	0.843	0.2111	0.2831
Multivar. conf. ($\|\mathbf{e}_{\mathrm{white}}\|_2^2$)	0.072	0.608	0.0062	0.0095
Fisher p-pool (224 dials)	0.109	0.643	0.0105	0.0183
Maximum 224-dial z	0.083	0.652	0.0093	0.0157
Hotelling T^2 (recalibrated)	0.283	0.816	0.2900	0.4831

19 streams × 3 seeds, causal means. System-matched rules match each stream's nominal panel alert rate after aggregation (Stouffer .095 F1, .0087 $\mathrm{FPR}_{\mathrm{hold}}$; 224-dial top-$k$ .070, .0077; Mahalanobis conf. .2789, CUSUM .3457, MEWMA .5862 $\mathrm{FPR}_{\mathrm{hold}}$).

VII. RESULTS AND DISCUSSION

A. Benchmark Results

Following the cohort hierarchy in Section V-B, Table IV reports the 33-stream development benchmark under retrospective buffered attribution (Panel (a)), causal streaming (Panel (b)), and per-dataset stratification (Panel (c)), while Table VI reports the five-block benchmark across all 118 untouched held-out streams and the 141-stream archive. On a separate 130-stream baseline screen, TS-JEPA reaches stream-mean Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082. The observed reduction in nominal false alarms with latent scoring is specific to these protocols; spherical normalization alone does not prove a distributional guarantee.

Operational trade-offs and configuration guidance: On the 33-stream benchmark, the 224-dial reference variant *Laminar-JEPA* edges out *Laminar-JEPA-Lite* on unweighted buffered Point-F1 (0.244 vs. 0.242) and causal ER (67.3% vs. 65.0%), whereas *Laminar-JEPA-Lite* halves buffered $\mathrm{FPR}_{\mathrm{hold}}$ (6.1% vs. 12.8%), and leads 7-dataset macro Point-F1 (0.171 vs. 0.133). In practice, we recommend *Laminar-JEPA-Lite* for streaming and buffered monitoring of abrupt and intermediate-onset faults and keep *Laminar-JEPA* for diagnostic ablation. Short horizons ($S \leq 4$) reduce reporting lag; raw-level variance or trend charts are needed only for single-sensor freezes or slow ramps.

Scope boundaries and controlled slow-ramp sweep: Because the encoder sees only the last $C = 256$ steps, a slow drift moves context and target summaries together, which explains the 0.000 Point-F1 on the 14 GHL slow-ramp streams. A controlled sweep maps this boundary. We injected linear ramps of amplitude $a \in \{0.5, 1, 2, 4\}\sigma$ and duration $L \in \{64, 256, 1024, 4096\}$ (15 events per cell in clean test segments, on multi-channel and single-channel subsets). Steep multi-channel ramps ($L = 64$, $a \geq 1\sigma$) are caught 86.7%–88.7% of the time at 46–54-step median delay, 5.7–7.7 percentage points above the chance rate of background alarm episodes. At $L = C$ the excess falls to 0.5–2.5 points, and for $L \geq 1024$ chance-corrected detection is indistinguishable from zero ($[-4.8, +0.8]$ points) for every JEPA detector. Fixed-window JEPAs thus target abrupt to medium-onset faults rather than gradual industrial drift.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table IV shows why 1D latent detectors degrade under causal streaming. Buffered attribution averages residuals across the overlapping predictions that cover each timestamp, whereas causal streaming scores a single right-edge step with no averaging, so one or two drifting coordinates can set the global SPOT threshold (Section I). A rate-matched Bernoulli alerter, $\mathrm{ER}_{\mathrm{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, catches only 28.1% of events at our $\leq 1.3\%$ causal FPR, but 60.0% at TimesNet's 24.3%.

C. Multi-Horizon Scaling Across $S \in \{1, 4, 16, 32, 64\}$ and Latency Trade-Offs

Because a target window of length S emits its causal decision at $\tau = t + S - 1$, $S = 64$ imposes a 63-step reporting lag. We retrain on five selected streams at $S \in \{1, 4, 16, 32, 64\}$ to examine the latency trade-off. At $S = 1$ all prefixes share one target step, and at $S = 4$ the shortest prefix is two steps, so these settings test the implemented detectors rather than distinct horizon views.

In Fig. 4(a), both *Laminar-JEPA* and *Laminar-JEPA-Lite* exceed 1D Multi-Horizon SPOT at every horizon with test FPR below 3.6%. At $S = 1$ and $S = 4$, *Laminar-JEPA* reaches 2.6× and 4.3× the scalar causal Point-F1 (0.037 and 0.050 vs. 0.014 and 0.012) with 77%–84% ER, while *Laminar-JEPA-Lite* reaches 2.1× and 3.1× (0.030 and 0.036) with 64%–82% ER; at $S = 4$, detection delay drops to 36.7–44.1 steps. Buffered Point-F1 peaks at 0.290 for *Laminar-JEPA* ($S = 32$) and 0.255 for *Laminar-JEPA-Lite* ($S = 64$) versus 0.182 for 1D SPOT. Five selected streams cannot establish a deployment rule across sampling rates or datasets.

D. Operating Budgets and Contextual vs. Point Anomalies

Fig. 4(b) and Table V examine detection across operating budgets and fault taxonomies:

1) *Autonomous vs. diagnostic operating budgets (Fig. 4(b), Table II(c)):* Along the causal recall–FPR frontier, the autonomous points of *Laminar-JEPA* and *Laminar-JEPA-Lite* occupy the $\leq$ 1.3% held-out FPR regime without test-label tuning. At post-hoc diagnostic test-FPR controls ($f^* \in \{1\%, 5\%\}$), *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% (63.1% unquenched) and 56.0% of events at $f^* = 1\%$ versus 39.5% for TS-JEPA, 21.7% for TimesNet, and 14.9% for TranAD.
2) *In-range contextual vs. point faults (Table V):* Under NASA's spacecraft taxonomy [4] (Panel (a)) and a 5σ amplitude split across the 19 development streams (Panel (b)), out-of-range point spikes ($> 5\sigma$) are caught by reconstructors and amplitude rules alike. In-range contextual faults instead change cross-channel phase or velocity while amplitudes stay normal. On NASA contextual sequences, *Laminar-JEPA-Lite* and *Laminar-JEPA* reach 87.5% and 77.5% autonomous ER at 1.3% and 2.2% FPR, versus 62.5% at 41.5% FPR for TimesNet.

TABLE IV
PRIMARY 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-horizon coordinate circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.128	**0.244**	**0.407**	**0.669**	432.9	0.013	0.144	**0.673**	249.9	0.004	0.000	0.133
Laminar-JEPA-Lite (128-Bit)	0.061	0.242	0.397	0.636	59.3	0.010	0.133	0.650	73.8	0.005	0.000	**0.171**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Multi-Horizon JEPA (1D)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Multi-Horizon JEPA (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.214	0.179	0.145	**0.229**	0.144	**.009/.061**	.012/.091	.030/.091	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.003/.004**	.005/.005	.004/.004	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.078	**0.191**	0.000	0.000	0.000	.009/.104	.008/.020	.004/.017	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.284**	0.210	0.161	0.158	0.151	.013/.151	**.004/.049**	.054/.095	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.357	0.375	**0.379**	0.289	0.167	.021/.223	.016/**.066**	**.014**/.135	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.010/.014	.001/**.000**	.008/.014	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.000	**0.240**	0.119	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), trained separately from Table II(a,b); best in **bold**, second underlined. In (c), n = streams, K = channels, E = events, π = anomaly rate, $F1_{all} = 2\pi/(1+\pi)$, Lam. = Laminar-JEPA, Lite = Laminar-JEPA-Lite. Unweighted Laminar-JEPA Δt reflects one late SWAN trigger (0.0002 recall; median Δt is 50.5 buffered / 45.8 causal, or 54.8 / 57.2 ex-SWAN). 95% stream-bootstrap CIs (n=19): Lite buffered FPR [.030, .103], F1 [.151, .335] and causal FPR [.006, .014], F1 [.055, .236]; Lam. buffered F1 [.150, .357] and causal FPR [.009, .018], F1 [.058, .257] (vs. Multi-Horizon 1D [.013, .097] and TimesNet causal FPR [.123, .375]).

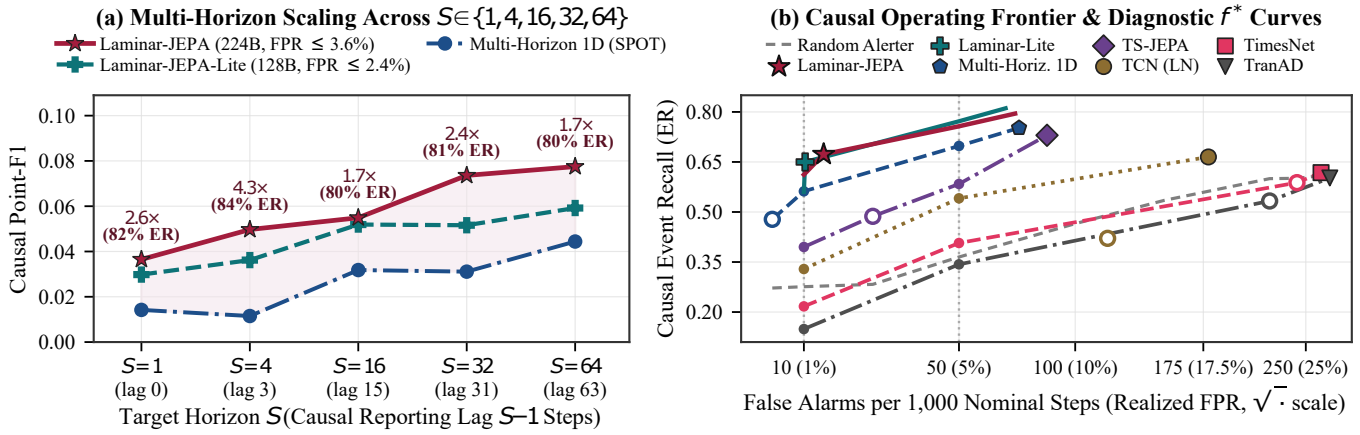


Fig. 4. Multi-horizon scaling ($S \in \{1, 4, 16, 32, 64\}$, causal reporting lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, n=5 streams) and causal operating frontier (n=19 streams). **(a)** Causal Point-F1 and event recall (ER) across S comparing *Laminar-JEPA (224B)*, *Laminar-JEPA-Lite (128B)*, and Multi-Horizon 1D SPOT (nominal FPR $\leq$ 3.6% across all S, vs. 2.5% $\rightarrow$ 21.3% for TimesNet). **(b)** Causal ER vs. realized false alarms ($\sqrt{\cdot}$ scale) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted guides), autonomous points (*Laminar-JEPA* star, *Laminar-JEPA-Lite* cross, 1D open circles), and quantiles $q^* \leq 0.05$.

E. Held-Out Generalization and Calibration Limits

Table VI and archive-wide audits establish three generalization and calibration findings:

1) *Independent generalization under five-block non-leakage (Table VI):* On the 118 untouched streams (three seeds), *Laminar-JEPA-Lite* raises causal ER from 52.5% to 63.8% over Multi-Horizon 1D, improving on all nine datasets (sign-flip $p = 0.0039$), and halves causal delay (70.8 vs. 133.5 steps). Causal Point-F1 rises from 0.076 to 0.079 ($p = 0.014$) and buffered Point-F1 from 0.143 to 0.156 ($p = 0.022$) at a median causal FPR_{hold} of 0.78%, and the panel beats the system-matched coordinate rules (Table VI). Without Exathlon ($n = 101$), *Laminar-JEPA-Lite* still leads buffered Point-F1 (0.148 vs. 0.139) and causal ER (50.6% $\rightarrow$ 58.8%), while Multi-Horizon 1D leads causal Point-F1 (0.073 vs. 0.060). Disjoint quenching barely moves causal FPR_{hold} of *Laminar-JEPA* (6.90% $\rightarrow$ 6.83%, sign-flip $p = 0.55$) and costs 3.5 points of ER, so *Laminar-JEPA-Lite* omits it.
2) *Autocorrelation, sample size, and cross-fitting:* Autocorrelation shrinks the effective calibration sample. Across the 19 development streams, the median Sokal integrated autocorrelation time τ_{int} is 2.51 for dial scores z_j, 1.88 for N_{edge}, 5.58 for N_{224} (max 206.0), and 6.75 for 1D scores. At median $N_{cal} = 408$, each quantile is set on a different statistic (z_j at $q_{0.985}$, N_{224} at $q_{0.995}$, N_{edge} at $q_{0.999}$), leaving median N_{eff} of 154.6, 67.1, and 195.9 and only 2.32, 0.34, and 0.20 expected exceedances; even $N_{eff} = 300$ would give 4.5, 1.5, and 0.3. Block-bootstrap ($L_{blk} \in \{1..128\}$) 95% CI half-widths span 12.7%–25.0%,

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION

(a) Model	NASA Contextual (n=5 seqs) Buf. PR	ER	FPR	NASA Point (n=8) f^*=.01	f^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	0.519	0.775	0.022	0.800	**0.960**	0.783	0.900
Laminar-JEPA-Lite	0.532	**0.875**	0.013	**0.840**	0.920	0.700	0.967
Multi-Horiz. 1D	**0.652**	0.750	**0.006**	0.820	0.920	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

(b) Model	19-Stream Contextual ($\|z\|\leq 5$) q^*=.05	f^*=.01	f^*=.05	19-Stream Point ($\|z\|>5$) q^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	**0.669**	0.448	**0.650**	0.835	0.644	0.795
Laminar-JEPA-Lite	0.655	0.376	0.567	**0.848**	0.581	**0.801**
Multi-Horiz. 1D	0.628	0.498	0.590	0.782	0.683	0.781
TS-JEPA	0.614	**0.520**	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Event recall (ER; autonomous vs. diagnostic f^*); best in **bold**, second underlined. (a) 10 SMAP/MSL streams by NASA taxonomy [4] (contextual: E-1, A-2, A-7, M-1, C-1 [2100, 2210]; point: A-1, P-3, D-3, D-14, D-15, C-1 [550, 750]). (b) n=19 non-GHL streams split into in-range ($\max|z| \leq 5\sigma$) vs. out-of-range ($\max|z| > 5\sigma$) causal events.

TABLE VI
DISJOINT FIVE-BLOCK HELD-OUT ARCHIVE BENCHMARK

(a) Untouched n=118	Pt-F1 ↑ Caus.	Buf.	PR-AUC ↑ Caus.	Buf.	ER ↑ Caus.	Buf.	FPR_{hold} ↓ Caus.	Buf.
TS-JEPA (1D)	.057	.134	.155	.207	.414	.459	**.042**	**.110**
Multi-Horizon JEPA (1D)	.076	.143	**.162**	**.230**	.525	.590	.044	.195
Laminar-JEPA (w/o Quench)	.075	.141	.143	.210	**.644**	.640	.069	.190
Laminar-JEPA (224B)	.072	.144	.143	.211	.609	.619	.068[§]	.190
Laminar-JEPA-Lite (128B)	**.079**[†]	**.156**[‡]	.145	.224	.638[¶]	**.674**	.070	.143
All Archive n=141 (12 **Datasets**)								
TS-JEPA (1D)	.055	.138	.167	.218	.414	.447	**.036**	**.102**
Multi-Horizon JEPA (1D)	.075	.146	**.174**	**.239**	.512	.573	.038	.181
Laminar-JEPA (224B)	.072	.150	.158	.233	.631	.628	.059	.175
Laminar-JEPA-Lite (128B)	**.076**	**.156**	.158	.235	**.636**	**.658**	.060	.124

(b) Held-out	n	Causal Pt-F1 ↑ Lite	Lam.	1D	Buffered Pt-F1 ↑ Lite	Lam.	1D	Causal FPR_{hold} ↓ Lite	Lam.	1D
SMAP	40	.083	.087	**.131**	.186	.180	**.192**	.087	.099	**.083**
SMD	18	.049	**.052**	.031	**.157**	.097	.096	.012	.019	**.003**
Exathlon	17	**.191**	.129	.092	**.204**	.166	.162	.180	.134	**.097**
Daphnet	15	**.072**	.064	.019	**.146**	.146	.101	.068	.077	**.007**
MSL	13	.033	**.055**	.052	.141	.151	**.182**	.012	.017	**.005**
OPPORT.	12	.009	.011	**.046**	.030	**.073**	.045	.021	.008	**.002**

Three seeds under $\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$; best **bold**, second underlined (1D = Multi-Horizon JEPA 1D). [†]Lite vs. 1D (n=118): causal F1 Δ=+0.003 (70/40 wins, p=0.014; vs. TS-JEPA +0.022, cluster CI [+0.005, +0.052], $p=1.8\times10^{-4}$); beats system-matched max-z (.062, $p=3\times10^{-5}$) and raw top-k (.073, p=0.0034). [‡]Buf. F1 Δ=+0.014 (p=0.022; vs. Lam. +0.012, p=0.0064, median buf. FPR_{hold} 4.79% vs. 8.70% 1D). [¶]Causal ER Δ=+0.113 vs. 1D (cluster CI [+0.051, +0.210], 9/9 sets, sign-flip p=0.0039; vs. ℓ_2^2 conf. +0.041, p=0.0041). [§]Disjoint $\mathbf{X}_q$ quench: causal FPR_{hold} $\Delta=-0.0007$ (p=0.010, sign-flip p=0.55), ER −0.035.

while floors $(5.0, 2.5)$ hold median FPR_{hold} to 0.36% and 0.00%. On $n = 118$, $N_{\text{cal}} < 250$ ($n = 12$) averages 16.0% causal FPR_{hold} for *Laminar-JEPA-Lite* (17.6% *Laminar-JEPA*) versus 1.05% (1.60%) for $N_{\text{cal}} \geq 2{,}500$ ($n = 22$); on short histories ($T_{\text{nom}} < 2{,}500$, $n = 18$), two-fold cross-fitting cuts causal FPR_{test} from 15.4% to 3.8% at 69.9% ER, while Beta quantiles and interquartile-range extrapolation remain exploratory.

3) *Regime drift vs. tail misfit via permutation decomposition:* Refitting thresholds across chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits is consistent with inter-block regime drift without ruling out parametric tail misfit: $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$ represents 71.3% of excess causal FPR in TS-JEPA and 78.3%–81.8% in reconstructors, versus 18.7% in the centered model. Within-checkpoint traces of *Laminar-JEPA* (n=6, $K \in \{9, 25, 55\}$) show that before tapping and centering, latent vectors sit off-center ($\|\bar{\boldsymbol{\mu}}_{\text{ctx}}\|_2 \in [4.73, 5.18]$ vs. $\sqrt{32} \approx 5.66$) and barely separate anomalies (divergence 0.06–0.16); tap fusion and barycenter subtraction restore divergence to 0.70–0.94 and cut mean coordinate D_{KS} from 0.140 to 0.096.

VIII. CONCLUSION AND FUTURE WORK

We evaluated a two-tier family of multi-horizon circuit-breaker architectures (*Laminar-JEPA-Lite* for deployment; *Laminar-JEPA* as a 224-dial reference) for abrupt and intermediate-onset telemetry faults across 33 development and 118 untouched held-out streams:

1) *Compact vs. reference panels:* On the 118 untouched streams, *Laminar-JEPA-Lite* raises causal ER from 52.5% to 63.8% over a single-score multi-horizon JEPA on all nine datasets ($p = 0.0039$) and improves causal and buffered Point-F1, with 35%–52% fewer parameters and 2.43× faster inference than *Laminar-JEPA*. On the 33-stream development benchmark the two tiers are close (0.244 vs. 0.242 buffered Point-F1 at 1.3% and 1.0% causal FPR_{hold}), and weight-space quenching gives no reliable benefit.
2) *Mechanisms and latency:* Matched-backbone controls show LayerNorm tail compression and barycenter drift reduction, and the five-stream horizon sweep keeps 2.1×–4.3× causal Point-F1 gains at $S \in \{1, 4\}$.
3) *Calibration, single-sensor, and ramp limits:* Autocorrelation leaves too few effective calibration samples to fix extreme quantiles precisely, though two-fold cross-fitting cuts short-stream ($T_{\text{nom}} < 2{,}500$) causal FPR_{test} from 15.4% to 3.8%. An optional channel branch helps injected single-sensor faults only at a much higher false-alarm rate, and excess causal detection vanishes on slow ramps once $L \geq 1024$ ($4C$); pairing raw-level trend charts and testing natural sensor failures remain future priorities.

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